LineSentry: Real-Time Machine Analytics Revolution

Transforming OEE, MTTR & MTBF Monitoring at Dawlance

Core Contributors



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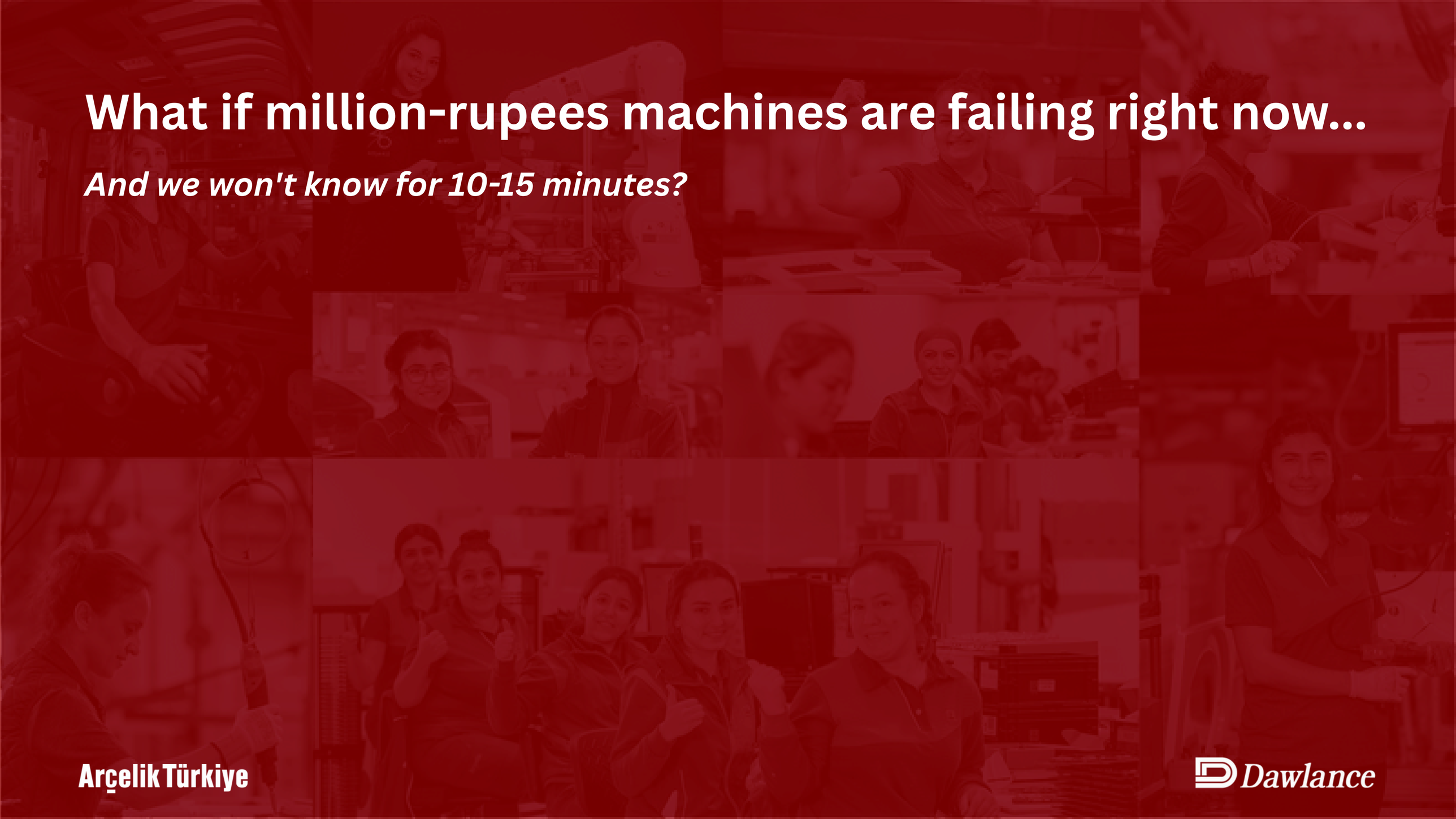
From €60K Vendor Lock-in to Open Intelligence

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What if million-rupees machines are failing right now...

And we won't know for 10-15 minutes?



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What if million-rupees machines are failing right now...

And we won't know for 10-15 minutes?

Every minute of delay costs us:



Lost productivity



Competitive disadvantage



Emergency repairs

In 15 minutes, you'll see how we change this forever

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The Crisis - Three Villains

Dawlance's Analytics Are Trapped



Real-Time Blindness

- SAP shows only historical data
- Breakdowns discovered after damage is done
- Manual calculation of analytics

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Vendor Captivity

- €60K for restricted access
- Zero customization control
- External dependency for everything

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Reactive Firefighting

- Manual alerts and escalations
- No predictive insights
- Crisis management every day

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Consequences of Inaction

What Happens If We Do Nothing?

Financial	*Operational*	Strategic
€60K-€70K in vendor costs	10-15 min delays per breakdown	Cannot identify root causes
Emergency repair expenses	Reactive maintenance culture	Falling behind competitors
Costly machine additions	Manual data entry errors	No optimization insights

Every day we wait is money lost and opportunities missed

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Sensitivity: Internal / Non-Personal Data

The Breakthrough Moment

What If We Could See Everything?

Imagine walking into the factory and instantly knowing:

- ☑ Which machine will fail next
- ☑ Which department is underperforming
- ☑ Exactly what needs fixing

Sensitivity: Internal / Non-Personal Data

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LineSentry

Real-time machine intelligence that puts YOU in control

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Sensitivity: Internal / Non-Personal Data

Solution Trinity

From Problems to Power

- Real-time blindness
- Vendor Captivity
- Reactive Operations



- Real-time awareness
+
- In-house software
+
- Proactive Operations
=

LineSentry

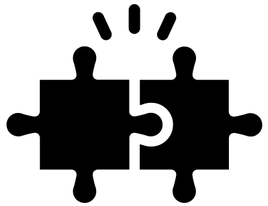
From Black Box to Open Control

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What is LineSentry?

Universal Machine Intelligence Platform



Universal Compatibility

Works with ANY machine via PLC integration



Live Analytics

Open-source flexibility at fraction of proprietary cost



Cost-Effective

Real-time OEE, MTTR, MTBF monitoring



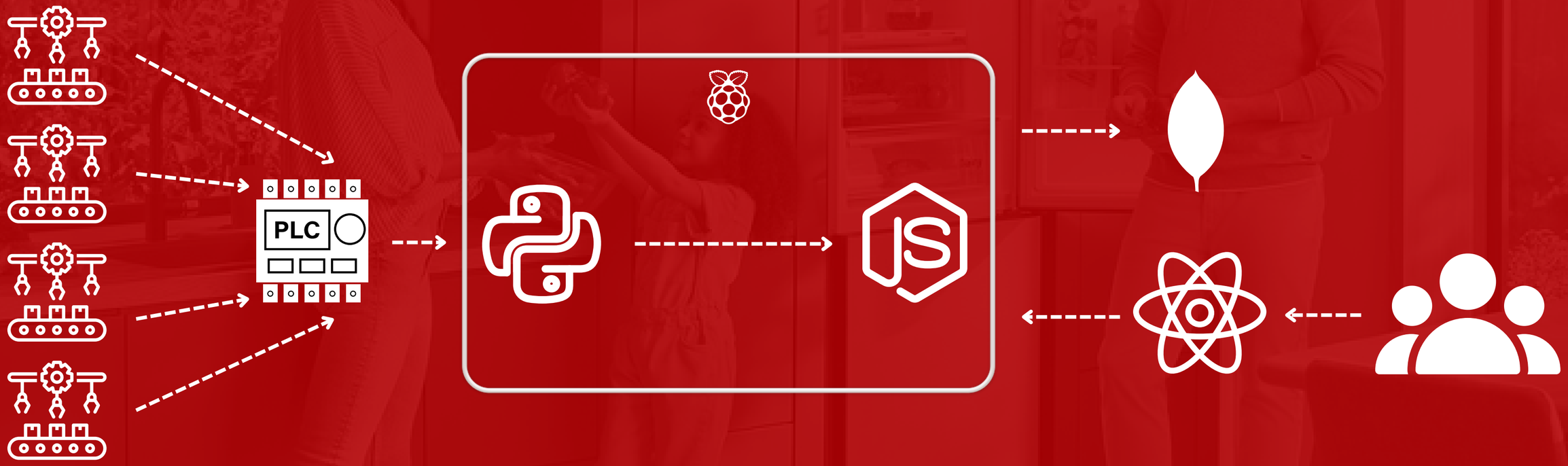
Instant Alerts

Automated breakdown notifications

Sensitivity: Internal / Non-Personal Data

The Technical Magic

How We Make Machines Talk?



LineSentry = Universal Translator

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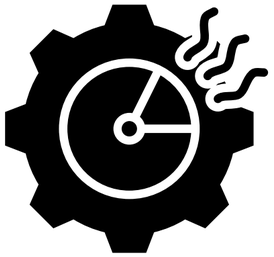
Analytics Engine Deep Dive

Making Math Exciting



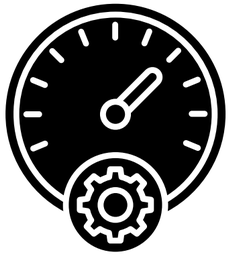
MTTR (Mean Time To Repair)

Total Downtime ÷ Number of Breakdowns



MTBF (Mean Time Between Failures)

Total Uptime ÷ Number of Breakdowns



OEE (Overall Equipment Effectiveness)

Availability × Performance × Quality

Sensitivity: Internal / Non-Personal Data

OEE Components Breakdown

The Intuition Behind the Numbers

Component	*Challenge*	LineSentry Solution
Availability	What counts as planned time?	Auto-classified via machine signals
Performance	How to determine ideal cycle time?	Dynamic baselines per mold
Quality	When to count rejected units?	Flexible hourly or shift-based

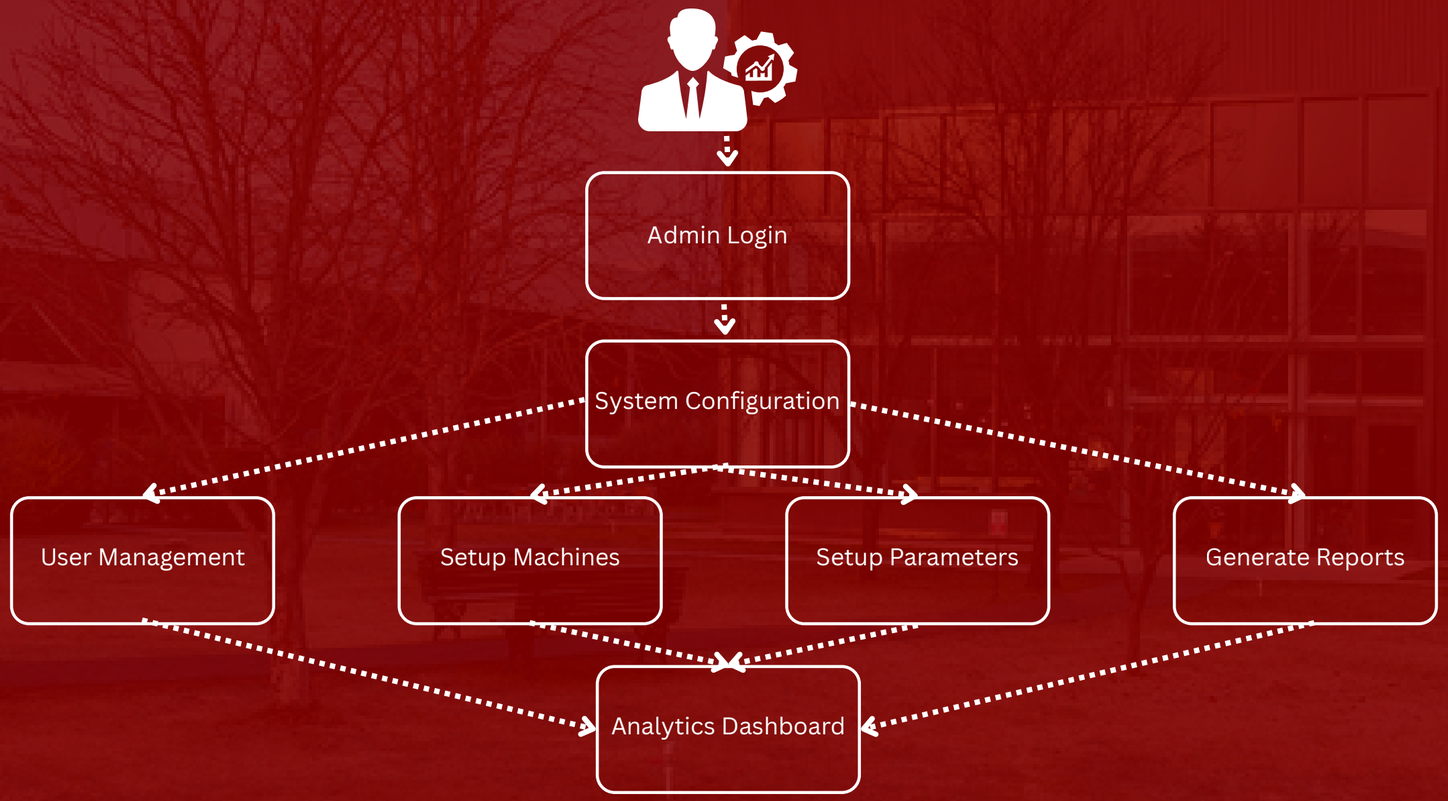
Accurate, automated, actionable insights

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Sensitivity: Internal / Non-Personal Data

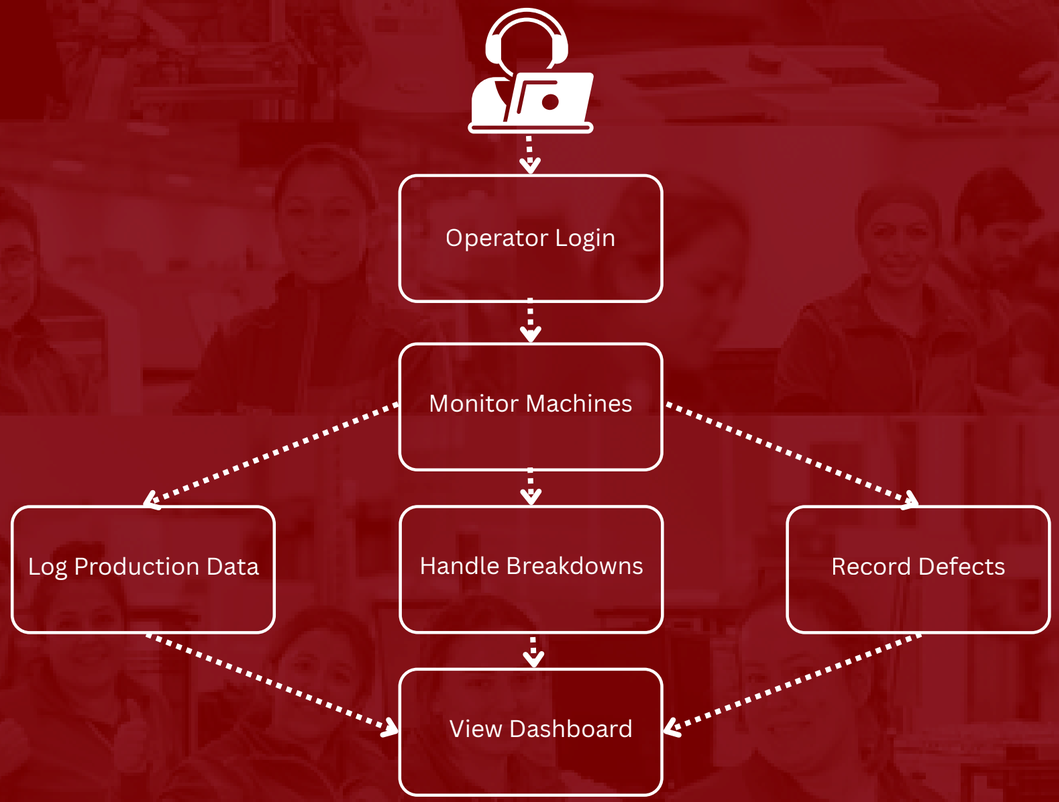
User Experience - Factory Manager

One Dashboard, Entire Factory



User Experience - Operator

From Reactive to Predictive



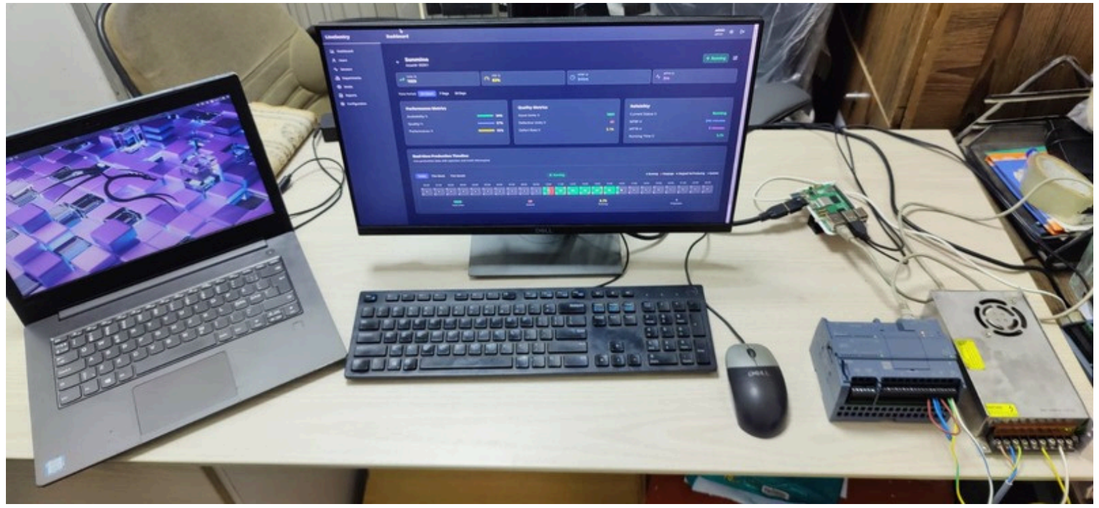
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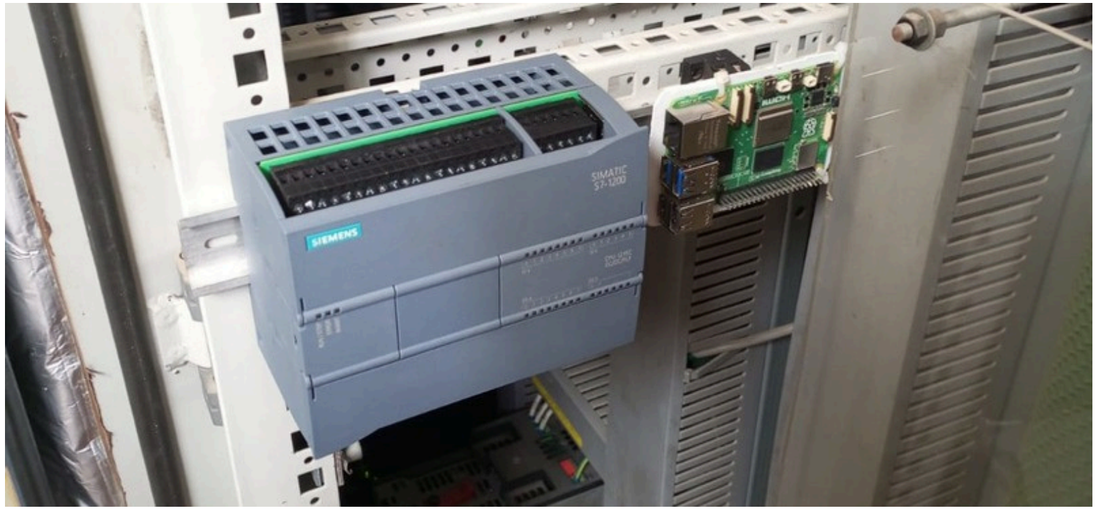
Proven Reliability

Lab Tests vs. Real-World Deployment



Foundation of Reliability

Scripted testing ensures algorithm accuracy before deployment



Proven in Battle

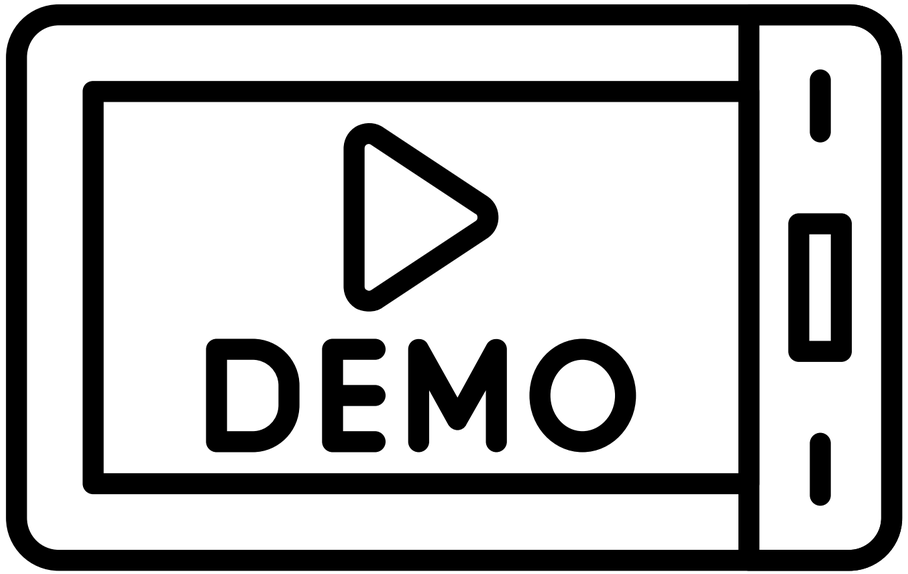
Real-time analytics operating on factory-floor machines

Sensitivity: Internal / Non-Personal Data

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Live Demo

See Real-Time in Action



Sensitivity: Internal / Non-Personal Data

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Business Impact - Immediate Wins

Why LineSentry Wins

Metric	Before	After LineSentry	Impact
Response Time	10-15 minutes	1-2 minutes	93% faster
Vendor Costs	€60K	€1K implementation	98% savings
Visibility	Historical reports	Real-time intelligence	Instant decisions
Machine Support	Vendor-limited	Universal compatibility	Unlimited growth

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Anticipated ROI

The Numbers Don't Lie

Year 1 Investment	Year 1 Savings
Implementation: ***€0.5K***	Avoided vendor fees: **€10K**
Training: ***€0.5K***	Reduced downtime: **€50K**
Infrastructure: ***€0.5K***	Efficiency gains: **€50K**
Total: €1.5K	**Total: €110K**

ROI: 2,200% in Year 1

Sensitivity: Internal / Non-Personal Data

System Architecture

Built for Scale and Reliability



- **Proven Tech Stack:** Node.js, React, MongoDB ensure stable performance
- **Enterprise Security:** JWT, CAPTCHA, XSS/CSRF protection
- **Real-Time PLC Integration:** Direct machine connectivity replaces legacy systems
- **Vendor-Independent Scalability:** Open hardware eliminates expensive proprietary lock-in

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Risk Mitigation

Addressing Concerns

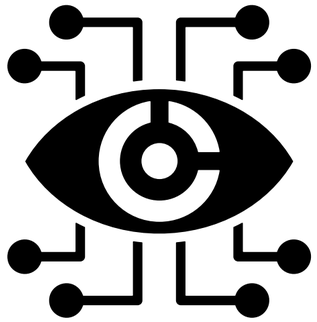
Concern	Mitigation Strategy
System Reliability	Redundant architecture
Data Security	Enterprise-grade encryption + RBAC
Staff Training	Intuitive UI + comprehensive training program
Integration Issues	Extensive testing + gradual rollout
Vendor Support	In-house expertise + open-source community

Sensitivity: Internal / Non-Personal Data

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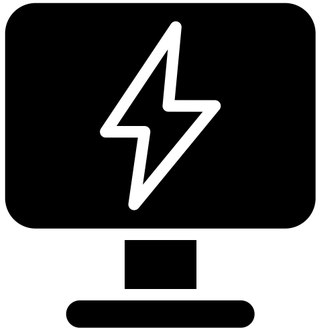
Future Vision

This Is Just the Beginning



Computer Vision

Automated defect detection



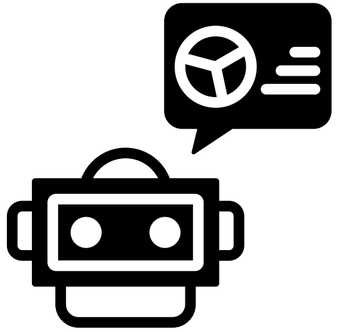
Energy Monitoring

Power consumption optimization



Biometric Security

Advanced access control



Predictive AI

Failure prediction algorithms

Sensitivity: Internal / Non-Personal Data

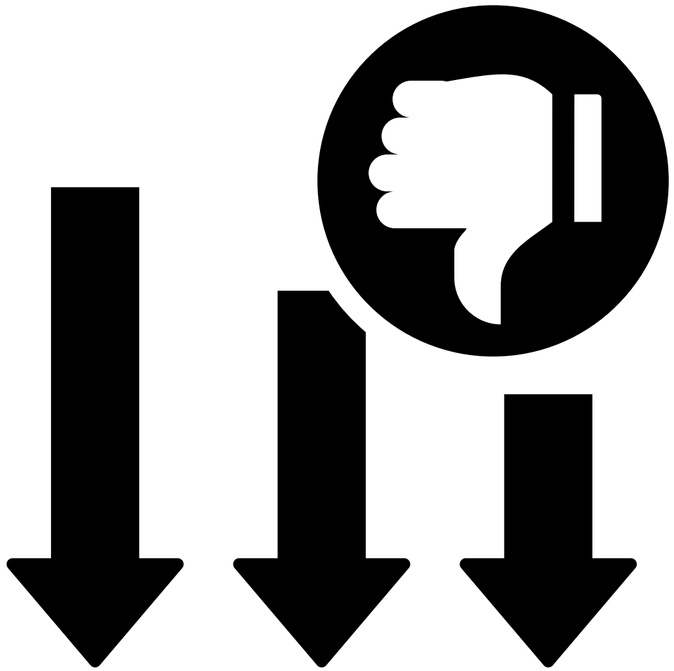
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The Moment of Decision

Can We Afford NOT to Do This?

Every day we wait:

- ⊗ Competitors get ahead
- ⊗ Vendor costs accumulate
- ⊗ Opportunities slip away
- ⊗ Problems multiply



LineSentry isn't just a system...

It's our competitive advantage

Sensitivity: Internal / Non-Personal Data

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Next Steps

From Vision to Reality



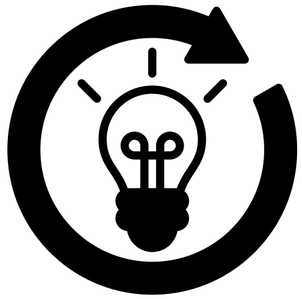
Approve pilot program

Sheet Metal Department deployment



30-day evaluation

Measure impact and ROI



Factory-wide rollout

Scale successful implementation



Industry leadership

Smart manufacturing pioneer

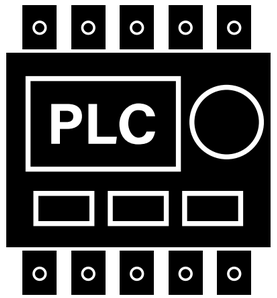
LineSentry operational in 60 days

Sensitivity: Internal / Non-Personal Data

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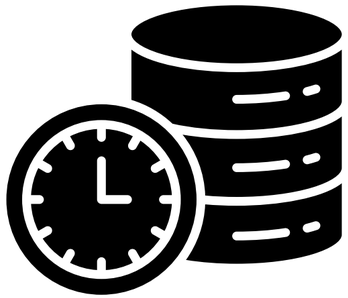
Key Takeaways as an Intern

What was learned along the journey



PLC programming

Reading/Writing data from/to PLC



Real-time Efficient Database

Optimizing performance for real-time applications



Business Analytics

Calculating machine analytics

Sensitivity: Internal / Non-Personal Data

Thank You & Questions

LineSentry Team

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Maintenance Head @ Dawlance



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